THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 05

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** Student A is correct	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$
2	**(b)** more than 7	Basic
3	**(d)** curd	Sour milk
4	**(b)** Incorrect — iodine is a lustrous non-metal	Exception
5	**(a)** Correct	Cu is more reactive than Ag
6	**(a)** 4	Two pairs
7	**(d)** propanol	–ol
8	**(d)** oxygen	Oxygen is released
9	**(d)** abscisic acid	ABA
10	**(d)** hibiscus	Both stamens and pistil
11	**(c)** Planaria	Cut pieces regrow
12	**(a)** phenotype	Observable
13	**(b)** alleles	T and t
14	**(d)** at 2F, same size	Real, inverted
15	**(a)** retina	Screen
16	**(a)** 1.6×10^{-19} C	Negative
17	**(b)** 3 units	1.5×2
18	**(c)** poles	Stronger field
19	**(c)** phytoplankton	Producers
20	**(a)** ecosystems	Biotic + abiotic

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	A yellow precipitate of lead iodide forms; the solution remains colourless. $Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$	1+1
22	Litmus (from lichen), turmeric, red cabbage, hydrangea/petunia flowers (any two)	1+1
23	Blood would not clot; a wound would keep bleeding.	2
24	It is a reflex action controlled by the spinal cord for protection.	2
25	False — Amoeba reproduces by binary fission; Plasmodium by multiple fission.	2
26	A checkerboard used to show the possible combinations of gametes and the offspring in a cross.	2
27	The centre of the sphere of which the reflecting surface is a part.	2
28	220 V; neutral is at (nearly) zero potential.	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Corrosion (rusting of iron), rancidity (stale chips), combustion (burning of fuels releasing energy)	1+1+1
30	Enrichment (concentration) of ore → conversion to oxide (roasting/calcination) → reduction to metal → refining	3
31	Impulse reaches the nerve ending, chemicals are released at the junction, muscle proteins change shape and arrangement causing contraction	3
32	Zygote divides to form embryo; ovule → seed; ovary → fruit; petals, sepals, stamens wither	3
33	Near point: minimum distance for clear vision (25 cm); far point: farthest point seen clearly (infinity)	3
34	Each segment produces circular field lines; near the centre arcs become straight and add up in one direction	3
35	(i) they are mild bases neutralising excess HCl (ii) to neutralise acids formed in the mouth (iii) to neutralise soil acidity	1+1+1

	OR: (i) $\mathrm{Fe} + 2\,\mathrm{HCl} \longrightarrow \mathrm{FeCl_2} + \mathrm{H_2}$ (ii) effervescence of $\mathrm{CO_2}$ (iii) solution stays colourless until neutralised, then turns pink with excess base	1+1+1
36	It is man-made and maintained by humans; biotic: fish, plants; abiotic: water, light, temperature, minerals	1+2
	OR: Enter food chains; biological magnification; kill useful organisms; soil and water pollution	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{10} + \frac{1}{15} = -\frac{1}{30}$ ⇒ $u = -30$ cm (ii) $m = v/u = 1/3$ (iii) virtual, erect, diminished image between F and O	2+1+2
	OR: (i) 15 cm (concave, $f = -15$) (ii) real, inverted, highly diminished (iii) at C, 30 cm (iv) $\frac{1}{v} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60}$ ⇒ $v = -60$ cm (v) yes, real and magnified	1+1+1+1+1
38	(i) addition: $\mathrm{CH_2{=}CH_2} + \mathrm{H_2} \xrightarrow{\mathrm{Ni}} \mathrm{CH_3CH_3}$; substitution: $\mathrm{CH_4} + \mathrm{Cl_2} \xrightarrow{\text{sunlight}} \mathrm{CH_3Cl} + \mathrm{HCl}$ (ii) substitution	4+1
	OR: (i) self-linking to form chains (ii) C–C bonds are strong and stable (iii) hexane, 2-methylpentane, cyclohexane	1+1+3
39	(i) parallel $= \frac{4\times12}{16} = 3\ \Omega$; total 4 Ω; $I = 1.5$ A (ii) voltage across parallel part $= 1.5 \times 3 = 4.5$ V; $I_{12} = 4.5/12 = 0.375$ A	3+2
	OR: (i) work done per unit charge between two points (ii) 6 J (iii) $1/(1.6 \times 10^{-19}) = 6.25 \times 10^{18}$	1+2+2
40	(i) taking in and utilising food (ii) ingestion by pseudopodia → food vacuole → digestion → absorption → egestion (iii) Paramecium uses cilia to move food to a specific spot (cytostome)	1+3+1
	OR: (i) Bowman's capsule with glomerulus, tubule, collecting duct (ii) filtration of blood in glomerulus; selective reabsorption of useful substances in tubules; remaining fluid is urine (iii) urea, formed in the liver	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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